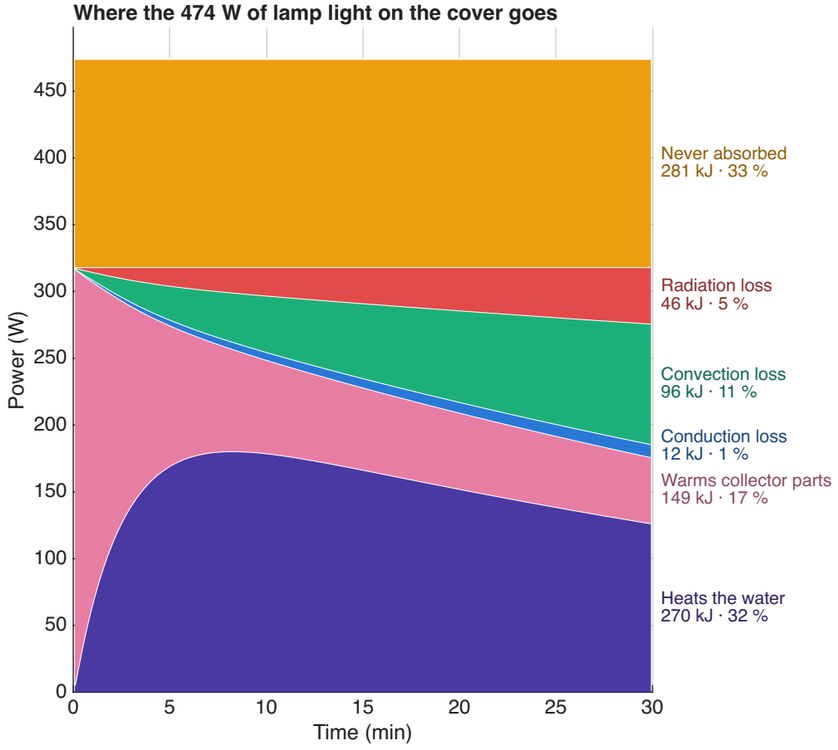
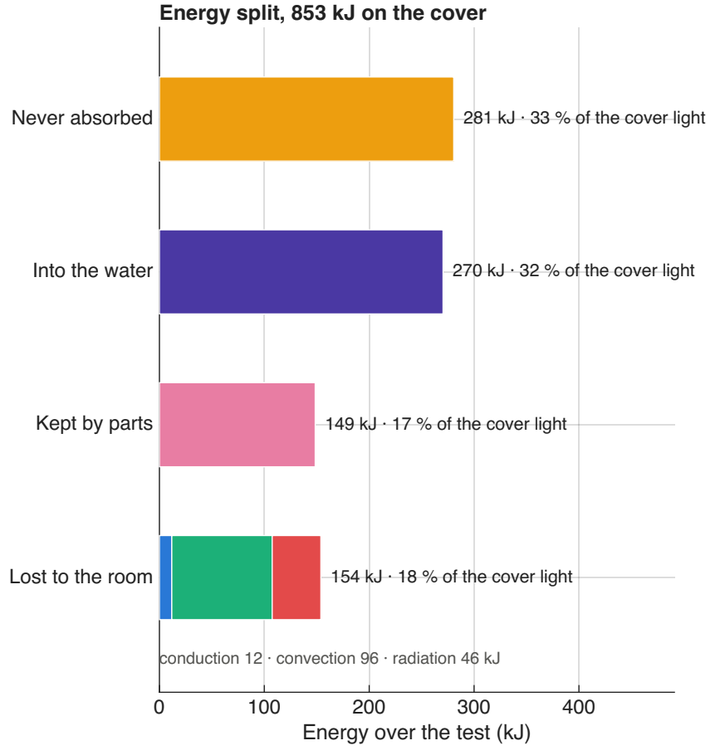
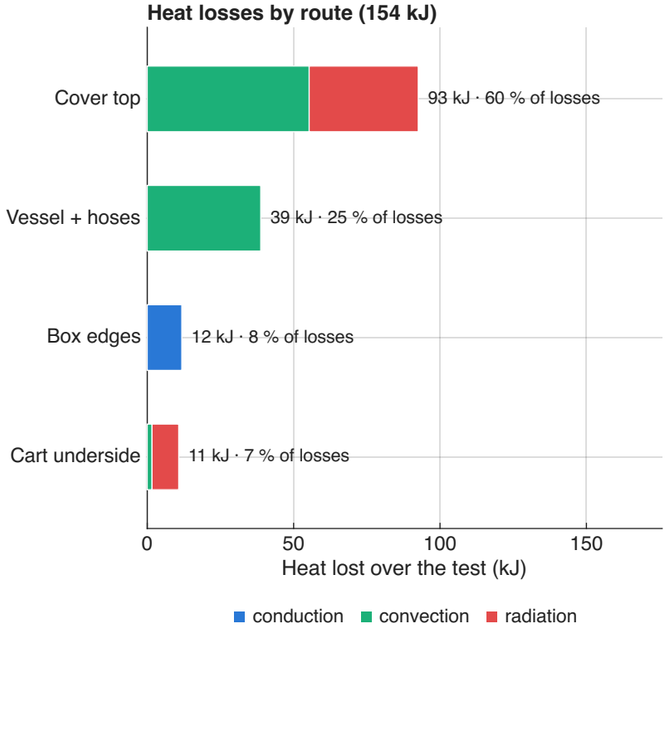

Where the 474 W of lamp light on the cover goes
Power (W)
Time (min)
0
50
100
150
200
250
300
350
400
450
0
5
10
15
20
25
30
Never absorbed
281 kJ · 33 %
Radiation loss
46 kJ · 5 %
Convection loss
96 kJ · 11 %
Conduction loss
12 kJ · 1 %
Warms collector parts
149 kJ · 17 %
Heats the water
270 kJ · 32 %


Energy split, 853 kJ on the cover
Never absorbed
281 kJ · 33 % of the cover light
Into the water
270 kJ · 32 % of the cover light
Kept by parts
149 kJ · 17 % of the cover light
Lost to the room
154 kJ · 18 % of the cover light
conduction 12 · convection 96 · radiation 46 kJ
0
100
200
300
400
Energy over the test (kJ)


Heat losses by route (154 kJ)
Cover top
93 kJ · 60 % of losses
Vessel + hoses
39 kJ · 25 % of losses
Box edges
12 kJ · 8 % of losses
Cart underside
11 kJ · 7 % of losses
0
50
100
150
Heat lost over the test (kJ)
conduction
convection
radiation